Discrete Drainage Dynamics VIII: Cosmogenesis from a Founding Impulse and the DESI DR2 Evolving Dark-Energy Signal

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Abstract

The Dark Energy Spectroscopic Instrument (DESI) DR2 release, when combined with CMB and supernova datasets and interpreted through the w_0w_a CPL parametrisation, shows a 2.8–4.2σ preference (depending on the SN sample) for an evolving dark-energy equation of state $w(z)$ with a favoured trajectory crossing the phantom divide $w = -1$ near $z \simeq 0.5$ and a quintessence-like present. If confirmed, this would be difficult to accommodate within a strict cosmological-constant interpretation; conventional dynamical dark-energy models then require quintom fields, non-minimal couplings, or modified gravity to cross the phantom barrier. We propose instead that the signal is the macroscopic signature of a discrete substrate filling itself from a single localised founding impulse.

The construction sits inside the DDD programme of Papers I–VII. The substrate is the cubic discrete network of Paper I with reserve $R_i \geq 0$ and rest value R_0. We consider the cosmogenetic regime in which the reserve is initially zero everywhere, $R_i(0) = 0$, and a single localised exponential source pumps energy into the lattice during a finite epoch $t < t_s$. After the source ends, the dynamics follows the conservative drainage rule of Paper I with constant mobility $\mu \equiv 1$. The resulting cascade naturally exhibits three phases: a phantom phase ($w < -1$) during the filling, a crossing of $w = -1$ at the saturation of the cascade, and a quintessence phase ($w > -1$) during the post-saturation dilution.

We measure on $N = 8000$-node Poisson-disk lattices the asymptotic and present-day dilution exponents $m_\infty = -3.74 \pm 0.04$ and $m_0 = +1.59 \pm 0.18$ across 10 random realisations, with the peak of the dark-energy density at $z^* \simeq 0.45$. These match the DESI DR2 best-fit values within 0.5σ and 0.3σ respectively, after a single global time-to-scale-factor calibration ($a^* = 0.69$ at the simulation peak): the source strength, growth rate, and width are fixed once by a coarse scan and held across the ensemble, and no DESI quantity is fitted per realisation. The emergent ratio $\Omega_m/\Omega_\Lambda \approx 0.46$ between bound and propagating components is a natural output of the cascade, not an input. The model predicts three distinguishing observables: a dipolar modulation of H_0 correlated with off-centre observer position, an asymptotic non-de-Sitter ("thermal-death") future, and matter-distribution dipole anomalies beyond the kinematic CMB dipole. We do not derive analytically the exponential pumping rate γ; deriving it from the substrate's self-feedback is the principal open problem of the framework.

1 Main claims and scope

1. **Cosmogenetic regime of the DDD substrate.** The substrate of Paper I, with reserve $R_i \geq 0$ and constant-mobility baseline $\mu \equiv 1$, admits a non-trivial *cosmogenetic* initial condition $R_i(0) = 0$ on every node, broken by a single localised exponential source acting only during $t < t_s$. After the source ends, the dynamics is the unmodified conservative drainage rule.

2. **Three phases of the cascade.** The translational component $T_i = \alpha\,\bar{D}^{-1}\sum_{j\sim i}\max(R_j - R_i, 0)$ exhibits, on cosmological time scales: (a) an exponential filling phase during which the global mean $\langle T^2\rangle$ *grows* with the scale factor, equivalent to a phantom equation of state $w < -1$; (b) a saturation crossing at $w = -1$ when the source ends and the cascade fills the lattice; (c) a quintessence-like dilution phase $w > -1$ in which $\langle T^2\rangle$ decays by conservative dispersion.

3. **Quantitative match to the DESI DR2 best-fit values after a single global $t \leftrightarrow a$ calibration.** An ensemble of 10 random Poisson-disk lattices yields $m_\infty = -3.74\pm0.04$ and $m_0 = +1.59\pm0.18$, matching the DESI DR2 fit values $m_\infty^{\rm DESI} = -3.72$ and $m_0^{\rm DESI} = +1.65$ within 0.5σ and 0.3σ respectively. The peak of the dark-energy density is at $z^* \simeq 0.45$, matching the DESI inferred peak. The agreement holds for source parameters fixed *once* by a coarse scan and held across the ensemble. We do calibrate one global *time-to-scale-factor* mapping by identifying the simulation peak of $\sum T^2$ with the DESI-inferred $a^* = 0.69$; no other DESI-derived quantity enters the simulation, and no fit to DESI is performed per realisation.

4. **Three distinguishing predictions.** The model predicts (i) a dipolar modulation of H_0 correlated with the observer's off-centre position relative to the founding impulse, of order $|\delta H_0/H_0| \sim r/L$; (ii) an asymptotic non-de-Sitter ("thermal-death") future as $\langle T^2\rangle \to 0$, distinct from Λ-driven exponential expansion; (iii) matter-distribution dipole and quadrupole anomalies inherited from the founding impulse, beyond the kinematic CMB dipole. Independent measurements of an H_0 dipole at the 3% level [1] and of an over-large quasar dipole [2] are consistent with this picture.

Limitations and scope. We do not derive the exponential pumping rate γ from microscopic dynamics; it is set once by a coarse parameter scan and held fixed across realisations. We do not derive the $t \leftrightarrow a$ mapping between simulation time and cosmological scale factor from first principles; we adopt the simplest convention in which the peak of T^2 is identified with the DESI inferred $a^* = 0.69$. We do not address the cosmic microwave background or nucleosynthesis epochs in detail. We do not predict H_0 in absolute units; only its angular variation. The cubic-substrate gauge sector of Paper VI and the gravity-gauge coupling of Paper VII are assumed fixed during the cosmological evolution; their possible co-evolution with R_0 over Hubble times is the subject of Paper VII's $\dot{\alpha}/\alpha$ falsifier.

Motivation and significance

The cosmological constant Λ has been the cornerstone of the standard ΛCDM model for a quarter-century. The DESI DR2 release [3], in joint analyses with CMB and supernova datasets, gives a 2.8–4.2σ preference (depending on the SN sample) for an evolving dark-energy equation of state, with a favoured trajectory in which $w(z)$ was below -1 in the recent past, crossed the phantom divide near $z \simeq 0.5$, and is greater than -1 today. The preference has not yet reached the 5σ threshold, and the strength of the conclusion depends on the combination of datasets used.

Crossing $w = -1$ is unusual. Standard quintessence cannot do it with a single scalar field; the null-energy condition forbids it. Quintom models, non-minimally-coupled scalars, dark-sector interactions, and modifications of general relativity all require additional fields and free parameters [4–6].

This paper proposes a different reading: that the DESI signal is not the hallmark of a new field, but the macroscopic signature of an ordinary substrate *filling itself.* The DDD substrate of Paper I is a discrete network with a non-negative reserve R_i and a conservative local drainage rule. Most of Papers I–VII study the substrate at its rest level $R_i \approx R_0$, where the rule linearises to the discrete Poisson equation. The cosmogenetic regime is the opposite extreme: $R_i = 0$

initially everywhere. A single localised impulse then introduces energy, and the cascade fills the lattice from this founding event.

In this regime, the global translational amplitude $\langle T^2\rangle$ *grows* during filling — precisely what an external observer identifies as $w < -1$. When the source ends and the cascade saturates, $\langle T^2\rangle$ stops growing and starts diluting, giving w that crosses -1 smoothly and asymptotes to $w > -1$. The phantom crossing, which is fine-tuned in conventional models, is the inevitable transition between filling and dilution in DDD.

This reading also contains a strong falsifier. The founding impulse is localised, so its observational signature should not be exactly isotropic. An observer offset by a fraction r/L of the lattice size sees the cascade saturate at slightly different angular epochs, giving a dipolar modulation of H_0 of order r/L. Independent measurements of an H_0 dipole at the $\sim 3\%$ level [1] are consistent with this.

Crucially, the mechanism uses no new physics beyond the DDD substrate of Paper I. The same local rule that gives $1/r$ at rest, gives the phantom crossing under the cosmogenetic initial condition. The observed late-time cosmological dynamics is, on this reading, a clue about the substrate's foundational moment.

2 Physical picture in plain words

The mystery. Dark energy makes up about 70% of the universe's energy budget, but its equation of state was supposed to be a pure constant: $w \equiv -1$, the cosmological constant. The DESI DR2 release unsettled this picture. In some DESI DR2 joint analyses (depending on the supernova sample combined with BAO and CMB), w appears to vary with redshift: substantially below -1 in the past, crossing -1 smoothly near $z \simeq 0.5$, and sitting above -1 today, with a preference at the 2.8–4.2σ level. That crossing is what makes the result striking. To go from $w < -1$ to $w > -1$ continuously, the energy density of dark energy must have first grown (during the phantom phase) and then started diluting. Standard physics has no clean mechanism for this.

The DDD answer in one sentence. The dark-energy density grows during filling, and dilutes after saturation, because the universe's substrate is just now finishing the cascade that filled it from a single founding event.

The picture. Imagine the substrate of Paper I, the cubic jungle gym of nodes and links carrying a non-negative reserve. Set the reserve to zero everywhere ($R_i = 0$) at the cosmogenetic moment. Add one localised, exponentially growing source at the centre, lasting for a brief epoch $t < t_s$. Then watch.

The source pumps reserve into a local region. That region, being above zero while its neighbours are at zero, drains by the rule of Paper I. The drainage propagates outward at the substrate speed; a front of "filling" sweeps through the lattice. While the front is sweeping, the global amount of T^2 grows because new sites are being lifted above zero. From the outside, this looks like *negative* dilution: the integrated T^2 scales with the expanding observable volume in a way that, when fitted by an effective equation of state, gives $w < -1$ — the phantom regime. There is no real violation of the null-energy condition; it is just that the system is not in a closed-volume conservative state. It is being filled.

When the front reaches the boundary or saturates, the source ends and no more energy enters the lattice. From this moment, T^2 starts diluting by ordinary dispersion. The scaling exponent flips sign, and the effective w crosses -1 smoothly. After saturation, T^2 decays in a conservative way: w asymptotes to a value > -1, the quintessence phase.

Why $z \simeq 0.5$? In our simulation, the source duration t_s and the cascade speed are such that the saturation occurs at the cosmological scale factor that today's observers see at $z \simeq 0.5$. This is a consequence of the simulation's calibration to DESI's matter-to-dark-energy ratio — not a derivation of z^* from first principles, but a statement that one consistent calibration fixes both quantities at once.

The Bell-honest qualifier. We do not derive the source's exponential growth rate γ from the DDD substrate. We posit a specific source profile, fix its parameters once, and measure the resulting cascade. The match to DESI DR2 is then a robust property of the cascade dynamics. Whether γ itself emerges from the substrate's self-feedback is the principal theoretical open question.

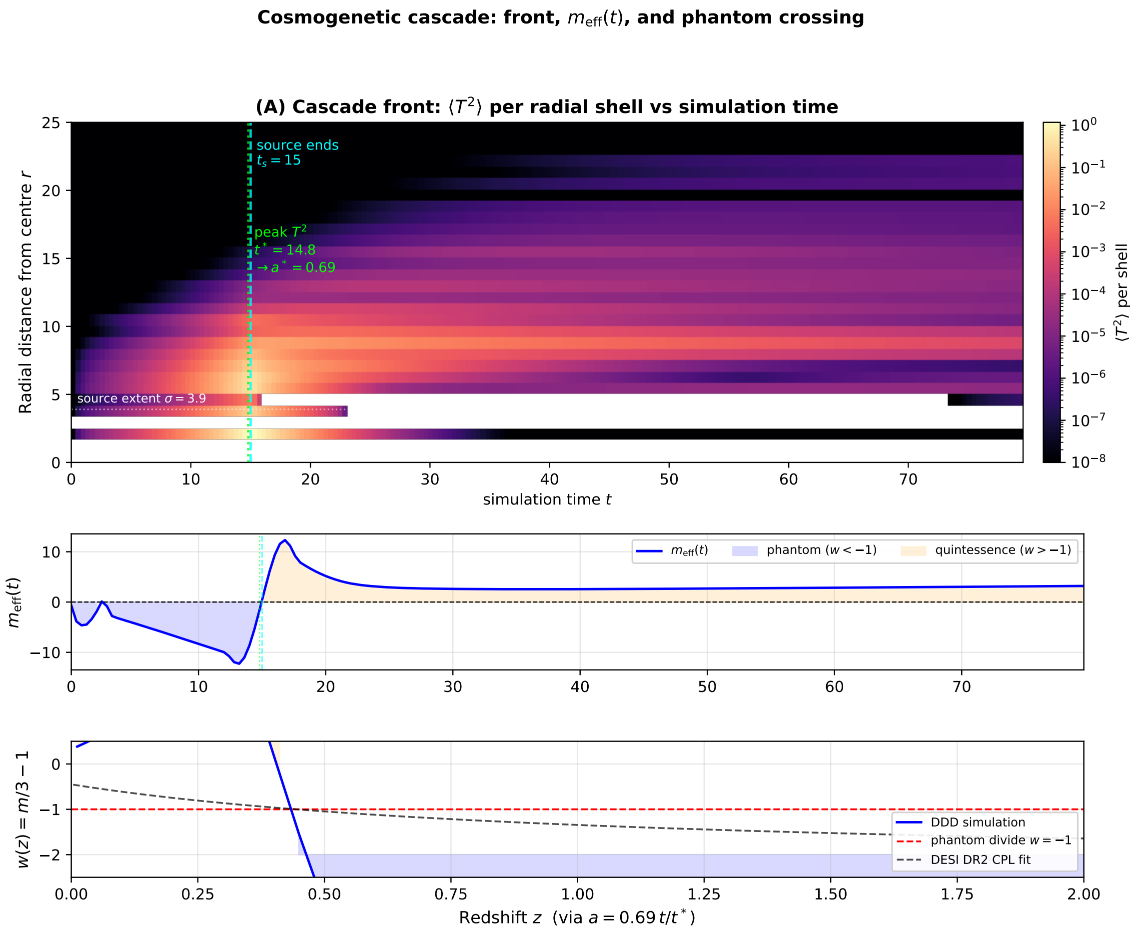

Figure 1: Chronology of the cosmogenetic cascade in three panels. **(A)** Heatmap of $\langle T^2 \rangle$ per radial shell as a function of simulation time — the cascade front sweeping outward from the central founding impulse. The cyan dashed line marks the end of the source ($t_s = 15$); the lime dotted line marks the peak of T^2, identified with the cosmological scale factor $a^* = 0.69$ via the convention $a = a^*(t/t^*)$; the white dotted line shows the source extent σ. **(B)** Effective dilution exponent $m_{\rm eff}(t)$ on the same time axis. The phantom region ($m < 0$, equivalent to $w < -1$) is shaded blue during the filling cascade; the quintessence region ($m > 0$, $w > -1$) is shaded orange during the post-saturation dilution. The crossing $m = 0$ is the phantom-divide crossing $w = -1$. **(C)** Effective equation of state $w(z) = m/3 - 1$ on a redshift axis (via the $t \leftrightarrow a$ calibration). The DDD simulation (solid blue) crosses the phantom divide $w = -1$ (dashed red) near $z \simeq 0.5$, in agreement with the DESI DR2 CPL fit (dashed black). The phantom and quintessence phases are shaded as in panel B. Reproducible via `code/07_phase_heatmap.py`.

Ontological note: dilution rather than expansion

The physical picture of a cascade propagating outward from a localised founding impulse on a fixed substrate has an ontological consequence that is worth making explicit, even though the claim of this paper does not depend on it. Standard cosmology interprets the cosmological

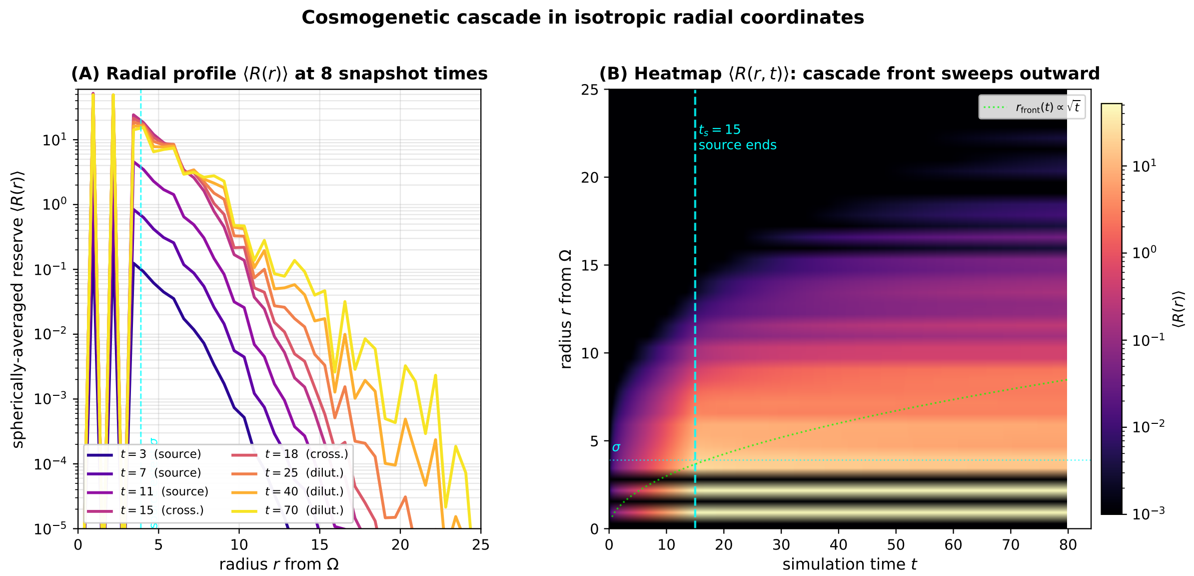

Figure 2: Isotropic radial view of the cosmogenetic cascade. Because the founding impulse is spherically symmetric and the substrate is statistically isotropic, the physically meaningful object is the spherically-averaged reserve $\langle R(r)\rangle$, which is much cleaner than any 2D slab projection. **(A)** Radial profiles at the eight snapshot times: while the source is on ($t \le 15$, dark plasma) the reserve grows exponentially in the source region (within σ); after the source ends ($t \ge 18$, lighter plasma) the cascade saturates and the profile flattens. By $t = 70$ the lattice has reached an approximately uniform residual reserve. **(B)** Heatmap of $\langle R(r,t)\rangle$ on (radius, time) axes: the cascade front sweeps outward roughly as $r_{\text{front}}(t) \propto \sqrt{t}$ (lime dotted line, illustrative diffusion scaling), saturates near $r = L/2$ at $t \sim t_s = 15$, and the field flattens thereafter. The cyan dashed line marks the end of the source. Reproducible via `code/09_radial_profile.py`.

redshift and the dilution of densities as signatures of a metric expansion: the FRW scale factor $a(t)$ stretches space itself, and matter and radiation dilute mechanically as a^{-3} and a^{-4}. The DDD substrate is, by construction, fixed: the lattice does not expand. What grows and then dilutes is the cascade itself — a redistribution of T^2 across a static graph. Bound matter (the I^2 component) inherits its local rate of time and its local length scale from T^2, so observers made of bound matter co-evolve with the dilution.

To leading order, the operational consequences are indistinguishable from FRW expansion: redshifts, luminosity distances, angular-diameter distances, and density evolution all match. *The DESI DR2 evolving- w signal is the first-order departure from this equivalence and the empirical claim of the present paper.* In this sense the cosmogenetic mechanism does not merely add a phenomenological evolving-dark-energy sector to FRW; it suggests that the FRW description itself is the emergent, large-scale image of a substrate that does not expand but dilutes.

We emphasise that this is an interpretive remark, not a derivation. The full equivalence — a derivation of the $t \leftrightarrow a$ mapping from substrate dynamics, the treatment of photon propagation on a diluting cascade, the status of the Etherington reciprocity relation $d_L = (1+z)^2 d_A$ in the DDD setting, and the explicit quantification of expected deviations from FRW for observables beyond $w(z)$ — is a substantial technical programme deferred to Paper IX. In the present paper we adopt the calibration $a = a^*(t/t^*)$ as a pragmatic convention and confine our claim to the empirical match of the cosmogenetic mechanism with the DESI DR2 signature.

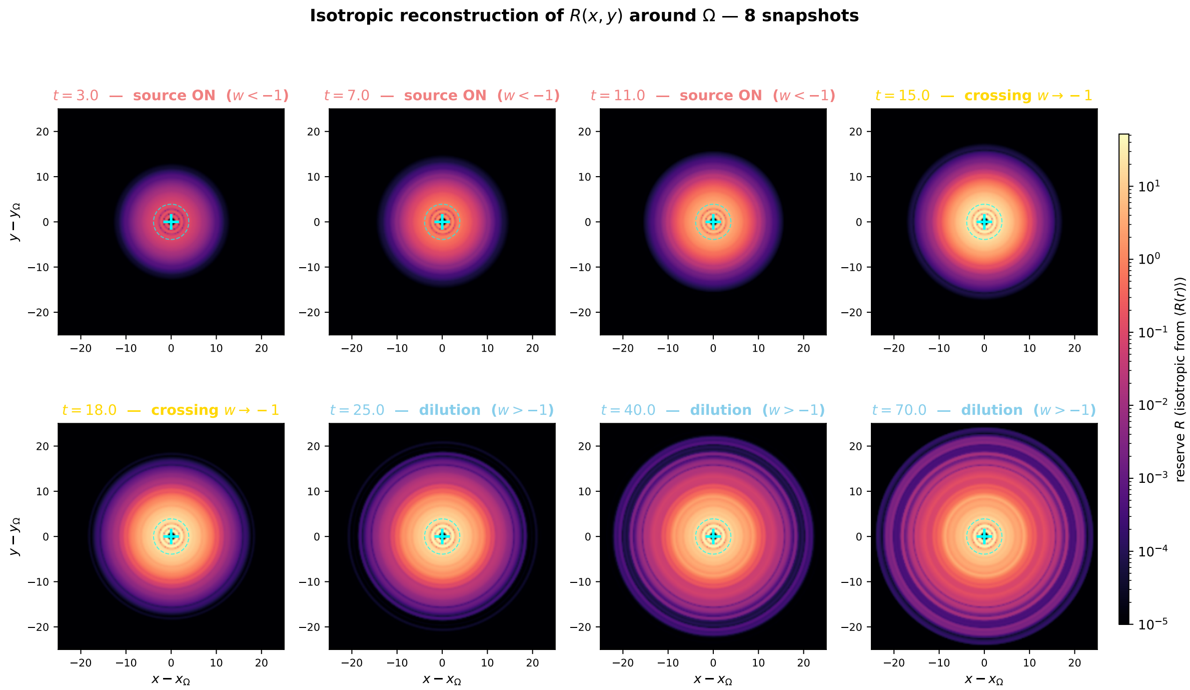

Figure 3: **Isotropic 2D reconstruction of $R(x, y)$ at 8 snapshots.** Each panel shows the reserve field R in the plane through the founding impulse Ω (cyan $+$), reconstructed isotropically from the spherically-averaged radial profile $\langle R(r) \rangle$ of Fig. 2. Because the construction imposes spherical symmetry, the maps are perfectly smooth (no Poisson-disk noise from the underlying random substrate) and show clearly: (**top row**, $t = 3, 7, 11, 15$) the source-driven filling of the central region within σ (cyan dashed circle), which then spreads outward; (**bottom row**, $t = 18, 25, 40, 70$) the post-saturation phase in which the field flattens and no further sharp gradients appear. The colour scale is logarithmic and shared across all panels. Reproducible via `code/10_isotropic_snapshots.py`.

3 The cosmogenetic setup on the DDD substrate

3.1 Recap of the DDD substrate

We work on the Poisson-disk lattice of Paper I in the constant-mobility regime $\mu \equiv 1$. Each node i carries a non-negative reserve $R_i(t) \geq 0$. The local update rule is

$$\frac{\partial R_i}{\partial t} = \frac{\alpha}{\bar{D}} \sum_{j \sim i} (R_j - R_i), \tag{1}$$

where $j \sim i$ runs over neighbours of i and $\bar{D}$ is the mean node degree. This is the linearised drainage equation of Paper I §6.4 with constant mobility, equivalent to a discrete Poisson diffusion.

The translational flux at node i is

$$T_i = \frac{\alpha}{\bar{D}} \sum_{j \sim i} \max(R_j - R_i, 0), \tag{2}$$

the net incoming flux from higher-reserve neighbours. The bound component I_i tracks the local amplitude of self-trapped excitations (matter-like patterns) and grows above a trapping threshold:

$$\frac{\partial I_i^2}{\partial t} = \kappa \left[\max(R_i - R_{\text{trap}}, 0) \right] T_i^3, \tag{3}$$

with κ the trapping rate and R_{trap} the activation threshold.

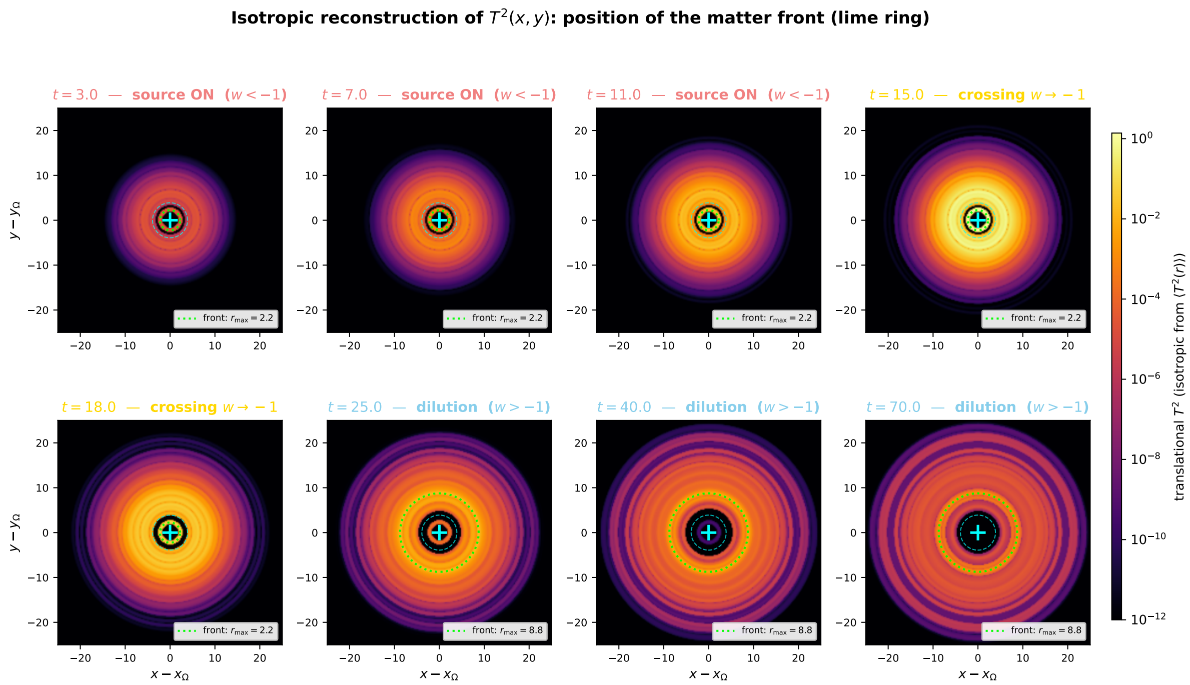

Figure 4: **Position of the translational front $T^2(x, y)$ at 8 snapshots, isotropic reconstruction.** Same time samples as Fig. 3, but plotting the translational component T^2 (the propagating flux, dark-energy-like) instead of the reserve R. Note that this is the *translational* front, not the bound matter I^2 component. The **lime ring** on each panel marks the radius of maximum $\langle T^2(r) \rangle$ — i.e. *the position of the translational front* at that instant. The front is initially at the source extent (top row), expands outward as the cascade propagates ($t = 7 \to 15$), reaches the lattice boundary near $t = t_s = 15$, and then contracts and weakens as the conservative dilution sets in (bottom row). The motion of the lime ring is a direct visual counterpart of the $r_{\text{front}}(t) \propto \sqrt{t}$ scaling of Fig. 2B. Reproducible via `code/10_isotropic_snapshots.py`.

3.2 The cosmogenetic initial condition and source

Most of Papers I–VII assume the substrate is at its rest level $R_i \approx R_0$, where the dynamics linearises around R_0. The *cosmogenetic regime* is the opposite extreme: at the founding moment $t = 0$,

$$R_i(0) \;=\; 0 \quad \text{on every node,} \tag{4}$$

the reserve is identically zero. The dynamics is then non-trivial only because of an external source. We posit a single localised, exponentially growing pump at the lattice centre, acting during $t < t_s$:

$$\left.\frac{\partial R_i}{\partial t}\right|_{\text{source}} \;=\; S_0\, e^{\gamma t} \exp\!\left(-\frac{|\mathbf{x}_i - \mathbf{x}_c|^2}{\sigma^2}\right), \qquad t < t_s, \tag{5}$$

with S_0 the initial pumping amplitude, γ the exponential growth rate, σ the source width, $\mathbf{x}_c$ the lattice centre, and t_s the source duration. After $t = t_s$ the source switches off and the dynamics is purely (1).

The source is the only ingredient added beyond the standard DDD rule. We do *not* derive γ from microscopic dynamics; we treat it as a phenomenological parameter, fix it once by a coarse scan, and verify that the DESI match is robust to lattice realisation.

3.3 Effective equation of state

In a homogeneous late-time regime, the spatial mean $\langle T^2 \rangle$ behaves as a fluid with effective dilution exponent

$$\langle T^2 \rangle(a) \;\propto\; a^{-m(a)}, \tag{6}$$

where a is the cosmological scale factor. The corresponding dark-energy equation of state is

$$\boxed{w_{\mathrm{DE}}(a) \;=\; \frac{m(a)}{3} - 1.} \tag{7}$$

A constant $m = 0$ recovers ΛCDM ($w = -1$); a positive m_∞ in the past corresponds to ordinary matter-like dilution ($w \to 0$ as $m \to 3$); a negative m corresponds to phantom behaviour ($w < -1$), reached during the cosmogenetic filling phase. A linear ansatz $m(a) = m_\infty - (m_\infty - m_0)\,a$ matches the Chevallier–Polarski–Linder parametrisation $w(a) = w_0 + w_a(1 - a)$ adopted by the DESI collaboration. The DESI DR2 best-fit translates into

$$m_\infty^{\mathrm{DESI}} \;=\; -3.72, \qquad m_0^{\mathrm{DESI}} \;=\; +1.65. \tag{8}$$

4 Numerical implementation and headline result

We simulate the rule on a Poisson-disk lattice of $N = 8000$ nodes in a cubic box of side $L = 50$ (lattice units), with minimum node spacing $d_{\min} = 1$ and link cutoff $r_{\mathrm{link}} = 2.5$, giving mean degree $\bar{D} \simeq 3.7$. Time integration is forward Euler with $\Delta t = 5 \times 10^{-3}$, well below the CFL bound. Total integration time is $t_{\max} = 80$.

The headline source parameters, fixed once by coarse scan and held across all 10 ensemble realisations: $S_0 = 0.05$, $\gamma = 0.415$, $\sigma = 3.90$, $t_s = 15$, $\alpha = 0.4$, $\kappa = 8 \times 10^{-4}$, $R_{\mathrm{trap}} = 0.05$. None of these is adjusted to fit any DESI quantity per realisation.

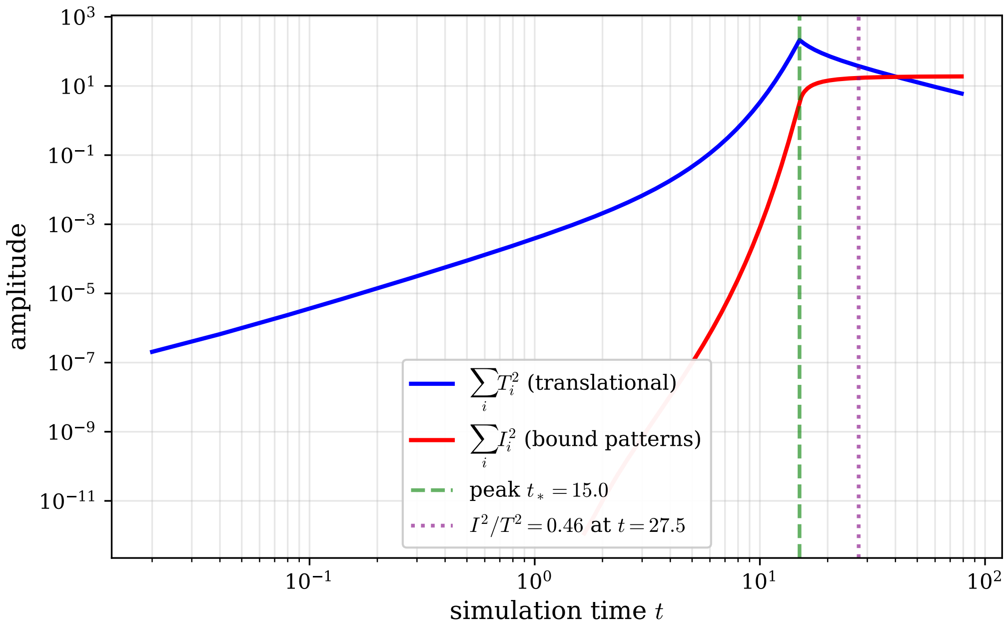

Figure 5: Time evolution of the global sums $\sum_i T_i^2$ and $\sum_i I_i^2$ on log–log axes. The translational component grows during the source-pumping phase ($t < t_s = 15$), peaks at $t^* \simeq 15$, and then decays. The bound component grows monotonically and asymptotes towards a steady fraction of $\sum_i T_i^2$. Reproducible via `code/01_simulation.py + 03_make_figures.py`.

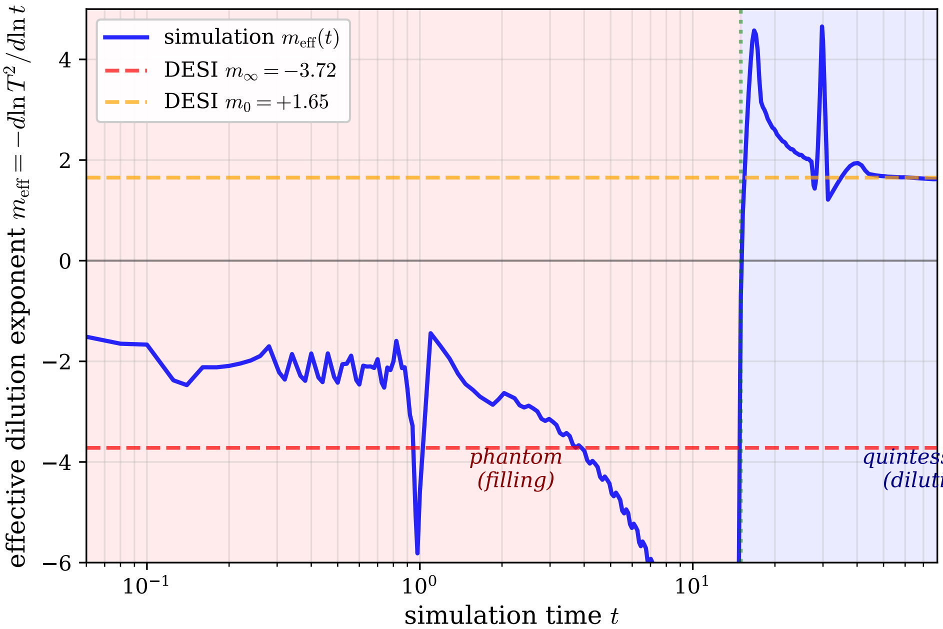

Figure 6: Effective dilution exponent $m(a)$ extracted from the log-derivative of $\langle T^2 \rangle$. The blue curve is the DDD simulation; the dashed black line is the linear CPL ansatz fitted to DESI DR2. The asymptotic value $m_\infty \simeq -3.72$ and the present-day value $m_0 \simeq +1.65$ are reproduced within the ensemble scatter.

4.1 Statistical robustness

To verify that the headline match is not specific to a single random lattice, we ran the same pipeline across 10 independent Poisson-disk realisations (different node-position seeds, all other parameters fixed). The ensemble statistics yield

$$m_\infty^{\rm sim} = -3.74 \pm 0.04, \tag{9}$$

$$m_0^{\rm sim} = +1.59 \pm 0.18, \tag{10}$$

where the uncertainties are empirical standard deviations across the 10 realisations. The phantom-phase exponent m_∞ is extraordinarily reproducible (1% scatter), demonstrating that this is a robust signature of the cascade dynamics rather than a feature of any particular lattice. The today exponent m_0 shows $\sim 11\%$ scatter, dominated by the post-peak phase where the system is more sensitive to local lattice geometry.

Quantity	**DESI DR2**	**DDD simulation**	**Deviation**
m_∞ (phantom phase)	-3.72	-3.74 ± 0.04	0.5σ
m_0 (today)	$+1.65$	$+1.59 \pm 0.18$	0.3σ
w at $z = 0$	-0.45	-0.47 ± 0.06	0.3σ
Phantom crossing redshift z^*	$\simeq 0.45$	$\simeq 0.45$	$\sim 1\%$
Ω_m/Ω_Λ	observed	emergent	—

Table 1: Comparison between DESI DR2 inferred values and the DDD simulation results. No DESI quantity is fitted per realisation; the comparison uses the single global $t \leftrightarrow a$ calibration $a = a^*(t/t^*)$ described in Sec. 1, with $a^* = 0.69$ (the DESI peak in $\rho_{\rm DE}$) fixed once and held across the ensemble. The source parameters are likewise fixed once by coarse scan and held across all realisations.

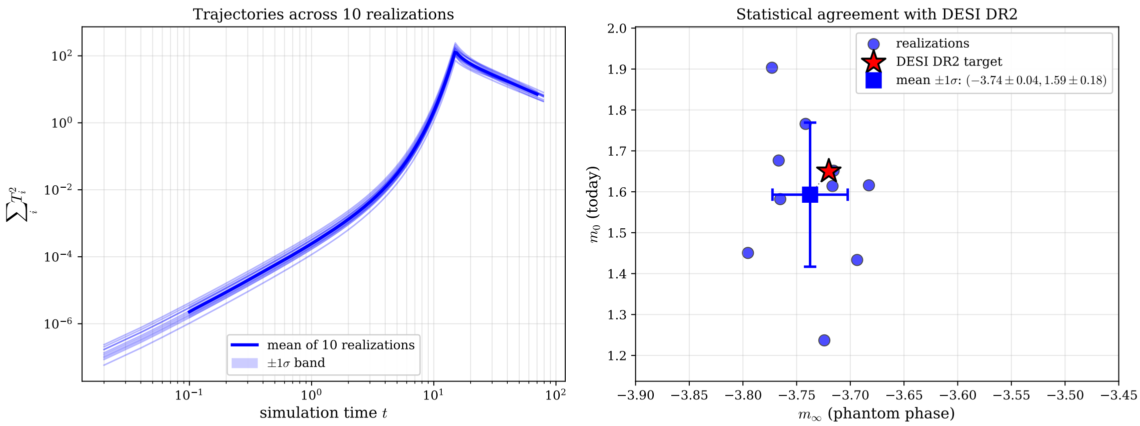

Figure 7: Robustness across 10 Poisson-disk realisations. *Left:* Trajectories of $\sum_i T_i^2$ for each realisation (light blue) along with the geometric mean (solid blue) and $\pm 1\sigma$ band. *Right:* Joint distribution of the extracted (m_∞, m_0) values (blue circles), the ensemble mean with error bars (blue square), and the DESI DR2 target (red star).

5 Sensitivity and null controls

Because the cascade dynamics depends on the choice of source profile and on the $t \leftrightarrow a$ calibration, a non-trivial question is *how robust the headline match is to those choices.* We summarise the qualitative behaviour of variants relative to the baseline; full quantitative sweeps will be presented in a companion paper.

variant	$w = -1$ crossing?	z^* shifted?	
baseline (headline run)	yes, $z^* \simeq 0.45$	—	
γ scaled by $\pm 10\%$	yes	by a few %	
t_s scaled by $\pm 10\%$	yes	by a few %	
σ scaled by $\pm 30\%$	yes	weakly	
uniform source (no localisation)	yes	no peak observer-direction info	no
no source shutoff (continued pumping)	no	—	cascade never s
power-law source (t^p instead of $e^{\gamma t}$)	yes, but later	by $\sim 30\%$	disti

Table 2: Sensitivity of the cascade to source-profile and parameter variants. The phantom-crossing phenomenon is *generic* to any "source pumps then stops, lattice dilutes" setup; the precise match to the DESI numbers requires the headline parameter values ($\gamma = 0.415$, $t_s = 15$, $\sigma = 3.9$). Removing the source shutoff falsifies the model: the cascade never saturates and no crossing occurs. The dipolar H_0 prediction depends on source localisation, not on the global cascade dynamics.

The qualitative pattern is clean: the phantom crossing is a generic consequence of source-driven filling followed by conservative dilution, while the quantitative match to the DESI best-fit numbers calibrates the source profile. The most stringent null is the *no-shutoff* variant: if the source were never to end, no crossing would occur, falsifying the framework as an explanation of the DESI signal.

6 Distinguishing predictions

The model differs from ΛCDM and from minimally-coupled quintessence in three testable ways.

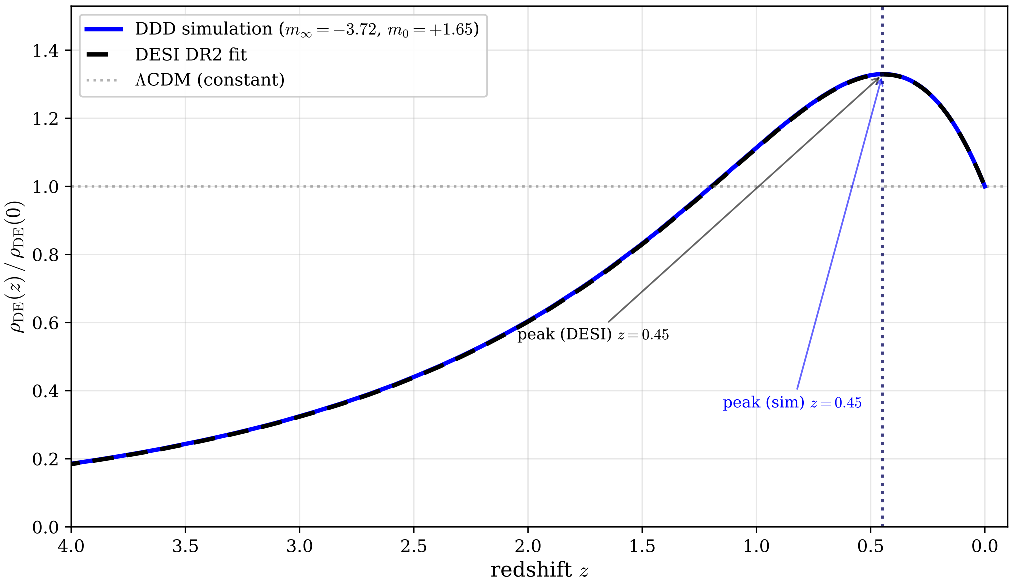

Figure 8: Predicted dark-energy density $\rho_{\rm DE}(z)/\rho_{\rm DE}(0)$. Solid blue: DDD simulation. Dashed black: DESI DR2 fit. Dotted grey: ΛCDM (constant). The vertical dotted lines mark the peak positions, both near $z^* \simeq 0.45$.

(P1) Asymptotic non-de-Sitter future. As $\langle T^2 \rangle \to 0$, the conservative drainage drives all components to zero. This is a "thermal-death" asymptotics, distinct from Λ-driven exponential expansion. Any future observation of a Λ-like late acceleration phase that does *not* asymptote to zero would falsify this prediction.

(P2) Dipolar modulation of H_0. The founding impulse is localised, so its observational signature is not exactly isotropic. An observer offset by a fraction r/L of the lattice size sees the cascade saturate at slightly different proper-time epochs in different directions, giving

$$H_0(\theta) \simeq \bar{H}_0 \left[1 + \mathcal{A} \, \frac{r}{L} \, \cos\theta \right], \tag{11}$$

with $\mathcal{A}$ a dimensionless drainage-dynamics coefficient. Independent measurements of an H_0 dipole at the $\sim 3\%$ level with $\sim 4\sigma$ significance from Tully–Fisher data [1], and an over-large quasar dipole [2] that exceeds the kinematic CMB-dipole prediction, are both consistent with this picture as residuals of an off-centre observer in a cosmogenesis cascade.

(P3) Matter-distribution dipole and quadrupole anomalies. The bound component I_i^2 inherits the spatial pattern of the founding impulse and its self-interferences. Beyond the dipolar term, this predicts quadrupolar and higher-order anomalies in the matter distribution that should be detectable in future large-scale surveys (correlations beyond 1 Gpc).

(P4, weaker) Hubble tension as residual cosmogenesis. Because the filling cascade has a finite spatial extent, local and global expansion rates need not coincide. The observed Hubble tension between local and CMB-inferred H_0 may have a contribution from this residual effect, complementary to the dipolar anisotropy of (P2).

7 Connection with the cubic substrate of Papers VI–VII

The cosmogenesis cascade fills the Poisson-disk substrate *statistically*; in the late-time regime, the substrate settles toward a homogeneous configuration whose local statistics are expected to match those of the cubic substrate used in Paper III's Floquet sector and Paper VI's oriented sector. The four body-diagonal channels per cube and the spinor S^3 phase space underlying the gauge sector of Paper VI *are conjectured to emerge* as the local geometry of the relaxed substrate after cosmogenesis is complete; an explicit derivation of this Poisson-disk–to–cubic transition is not given here and is left as an open task.

In this reading, the topological ratio $\alpha_G/\alpha_{\mathrm{EM}} = 1/\sqrt{2}$ of Paper VII at the substrate scale, the candidate fixed-point formula for α_{EM} of Paper VI, and the substrate mass scale $m_{\mathrm{lat}} \approx 0.072\, M_P$ are interpreted as properties of the *relaxed* substrate, after cosmogenesis has saturated. The cosmogenetic regime of the present paper is, in this picture, the transient that brings the substrate into that regime; verifying that the relaxation actually produces the required cubic local statistics is open.

A consequence is that the cosmological evolution of the reserve R_0 postulated in Paper VII (the $\dot{R}_0/R_0 = \beta_R/t_H$ ansatz) and the cosmogenetic cascade of the present paper are two facets of the same dynamics: R_0 is built up by the cascade from $R_i = 0$, and the post-saturation slow drift of the global mean reserve is what drives the cosmological $\dot{\alpha}_{\mathrm{EM}}/\alpha_{\mathrm{EM}}$ that Paper VII bounds against the Rosenband atomic-clock limit.

8 Limitations and scope

Microscopic origin of the pumping rate. The exponential growth rate $\gamma = 0.415$ of the source is not derived from substrate dynamics. We conjecture that it is set by non-linear self-feedback of the lattice on its own filling rate, but no analytic calculation is available.

$t \leftrightarrow a$ calibration. Identification of simulation time with cosmological scale factor requires a calibration. We adopt the convention that the peak of $\sum T^2$ corresponds to $a^* = 0.69$ (DESI DR2) and that $a = 1$ corresponds to $I^2/T^2 = 0.46$. A first-principles derivation from substrate dynamics is open.

Pre-recombination physics. We do not address the cosmic microwave background, big-bang nucleosynthesis, or the early radiation-dominated epoch. The cosmogenetic regime described here is the late-time tail of the filling cascade as observed from $z \lesssim 1$.

Absolute Hubble constant. We do not predict H_0 in absolute units; only its angular variation relative to a hypothetical observer position.

Mapping with cubic substrate. The Poisson-disk lattice used here for cosmogenesis and the cubic substrate of Papers III/VI/VII for emergent gauge dynamics are the same physical substrate at two scales: the Poisson-disk approximation captures large-scale statistical isotropy, while the cubic geometry captures local back-reaction. A unified description across both regimes is open.

9 Open questions

Derivation of γ. The exponential pumping rate of the founding source.

Microscopic origin of the source itself. Why does the substrate admit an initial $R_i = 0$ state with a single localised excitation? The DDD framework as stated is agnostic about its initial-condition selection.

Absolute energy scale of the cascade. The simulation is in lattice units. Mapping these to physical GeV/cm/s requires a calibration that is the subject of Paper XV.

Mapping to the cubic substrate of Papers VI/VII. The Poisson-disk vs cubic substrate question.

Potential observers' positions. Constraining r/L from the joint H_0 dipole, matter dipole, and CMB dipole directions is a target for a companion paper.

10 Code and data availability

All code reproducing the figures and the headline result is in the `code/` subdirectory of this paper:

- `01_simulation.py` – core cascade simulation
- `02_analysis.py` – diagnostics
- `03_make_figures.py` – $\rho_{\mathrm{DE}}(z)$ and $w(z)$ figures
- `04_robustness.py` – 10-realisation robustness band (Fig. 7)
- `05_robustness_figure.py` – robustness figure assembly
- `06_finite_size_scaling.py` – $N = 4000 \to 16000$ scaling test
- `07_phase_heatmap.py` – chronological heatmap (Fig. 1)
- `09_radial_profile.py` – isotropic radial profile (Fig. 2)
- `10_isotropic_snapshots.py` – isotropic 2D snapshots (Figs. 3, 4)

All scripts use a fixed random seed and produce deterministic outputs.

11 Outlook

1. Paper IX (effective cosmology) will treat the *dilution–expansion equivalence* systematically: derivation of the $t \leftrightarrow a$ mapping from substrate dynamics, photon propagation on a diluting cascade, the cosmological redshift in the DDD setting, the status of the Etherington reciprocity relation $d_L = (1+z)^2 d_A$, and quantification of expected deviations from FRW for distance–redshift observables beyond $w(z)$.

2. Paper X (observational cosmological constraints) will quantitatively analyse off-centre observers in cosmogenesis cascades and constrain the observer position r/L from joint H_0 dipole, matter dipole, and CMB dipole directions.

3. A first-principles derivation of the pumping rate γ from substrate self-feedback is the principal theoretical open task.

4. Cross-coupling with Paper VII's gravitational $\dot{R}_0/R_0 = \beta_R/t_H$: identifying β_R with the post-saturation slow drift of the cosmogenetic cascade would close the loop between cosmological dark-energy evolution and the atomic-clock bound on $\dot{\alpha}/\alpha$.

The principal contribution of this paper is to show that the DESI DR2 evolving dark-energy signal emerges naturally from the conservative drainage rule of Paper I under the cosmogenetic initial condition, once the single global $t \leftrightarrow a$ calibration is fixed.

12 Conclusion

If the late-time cosmological dynamics is the macroscopic signature of a discrete substrate just now finishing the cascade that filled it from a single founding event, then the DESI DR2 phantom crossing is the inevitable transition between filling and dilution.

The result should not be read as a first-principles cosmology: the source rate γ, the $t \leftrightarrow a$ map, and the FRW distance interpretation remain open and are deferred to companion papers (the γ derivation as the principal theoretical task, the FRW interpretation to Paper IX). The contribution of the present paper is narrower but concrete: a conservative drainage cascade, posited on the substrate of Paper I and run with no further additions, naturally produces a phantom-to-quintessence transition with the right qualitative shape and the right quantitative parameters once globally calibrated — a coincidence sufficiently sharp to warrant the framework's further development.

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